

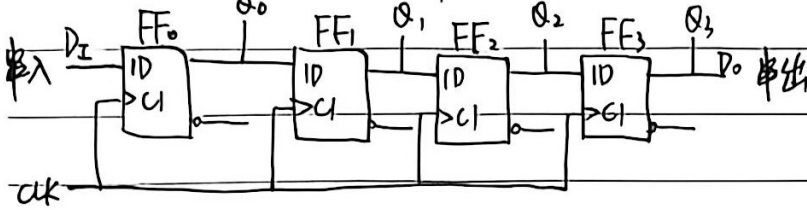
时序电路

No.

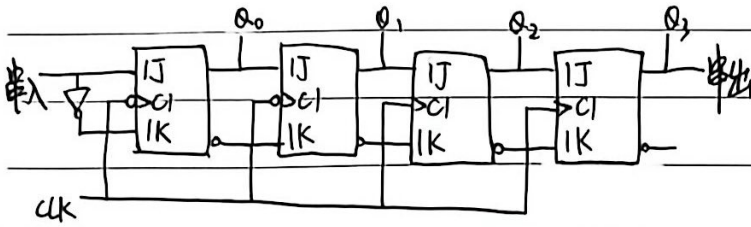
Date. / /

移位寄存器

1. 4位 串入 串出

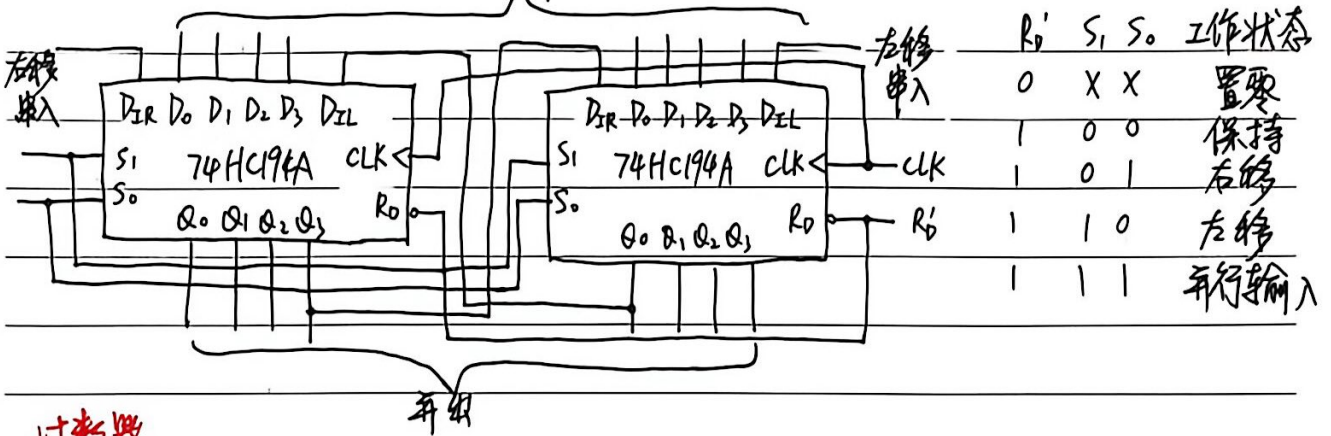


CLK	C_1	Q_0	Q_1	Q_2	Q_3
0	0	0	0	0	0
1	1	1	0	0	0
2	0	0	1	0	0
3	1	1	0	1	0
4	1	1	1	0	1



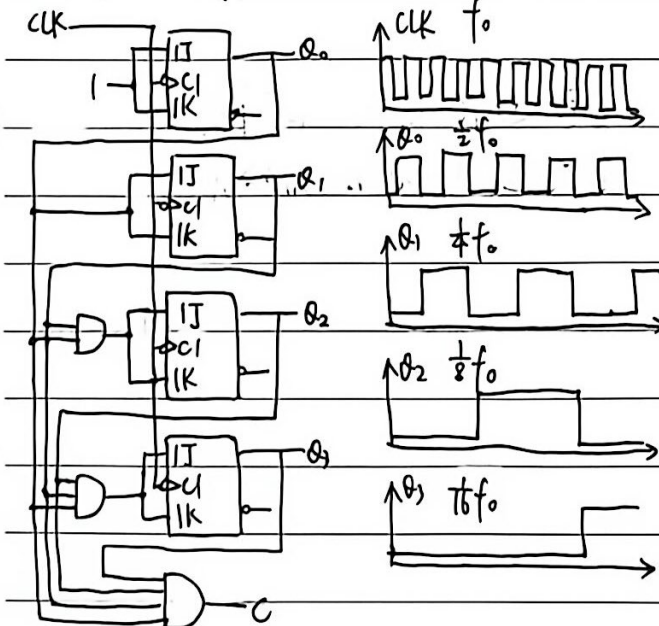
串入 → 串出
串入 → 串出
串入 → 串出

2. 双向移位寄存器 74HC194A



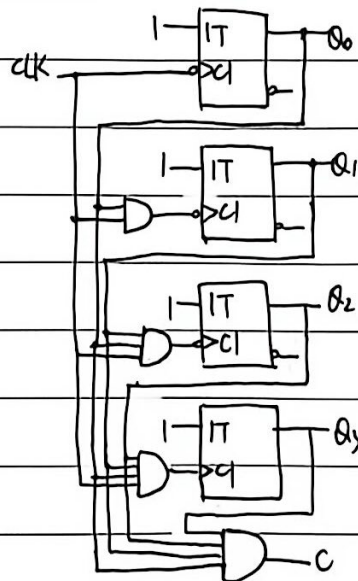
计数器

1. 同步二进制加法



另:

减法



$$T_0 = T \cdot Q_j$$

$$C = Q_0 Q_1 Q_2 Q_3$$

$$clk_i = CLK \cdot T \cdot Q_j$$

$$C = Q_0 Q_1 Q_2 Q_3$$

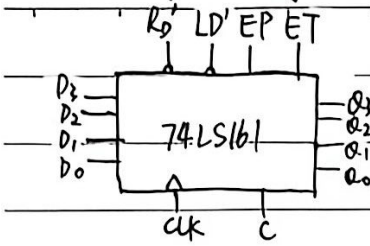


CS 扫描全能王

3亿人都在用的扫描App

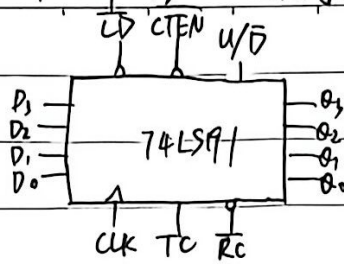
No.

① 4位同步二进制加法



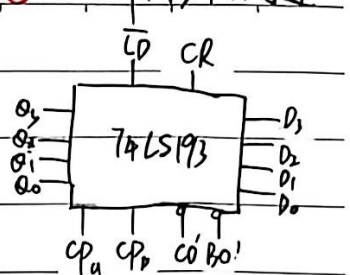
CLK	R0'	LD'	EP	ET	工作状态
X	0	X	X	X	置数
↑	1	0	X	X	预置数
X	1	1	0	1	保持
X	1	1	X	0	保持(但C=0)
↑	1	1	1	1	计数

② 单时钟同步十六进制可逆



U/D = 1 加 0 减
CTEN (G) 计数允许
TC 计数输出
RC 进/借位输出

③ 双时钟同步十六可逆



CO 非同步进位输出
BO 非...借...

2. 同步十进制

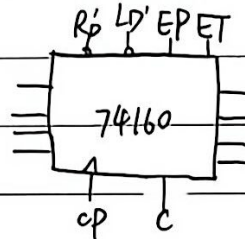
$$\begin{cases} T_0 = 1 \\ T_1 = Q_0 Q_3 \\ T_2 = Q_0 Q_1 \\ T_3 = Q_0 Q_1 Q_2 + Q_0 Q_3 \end{cases}$$

加法

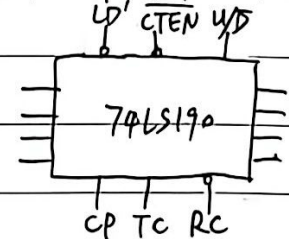
$$\begin{cases} T_0 = 1 \\ T_1 = Q_0' (Q_1' Q_2' Q_3)' \\ T_2 = Q_0' Q_1' (Q_1' Q_2' Q_3)' \\ T_3 = Q_0' Q_1' Q_2' \end{cases}$$

减法

① 同步十进制加

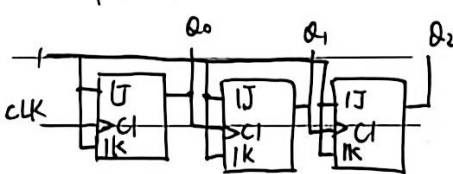


② 单时钟同步十进制可逆

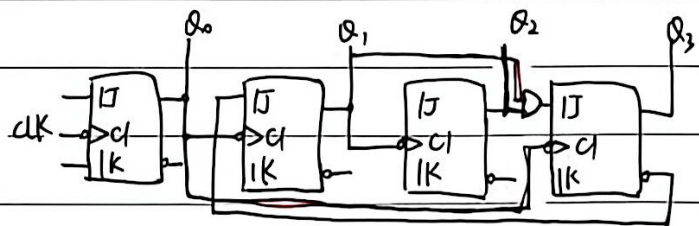


3. 异步计数器

从1001返回0000



① 二进制



② 十进制

任意进制计数器 (用N构成M)

置数: 延迟 R0' 由 S_{m-1} 译出
置数: 立刻 LD' 由 S_{int} 译出



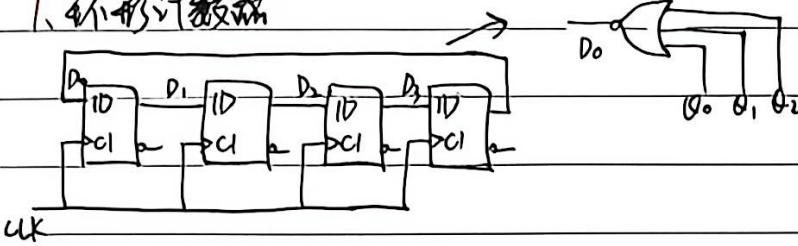
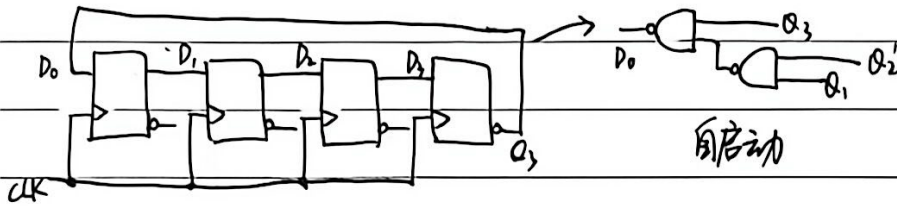
CS 扫描全能王

3亿人都在用的扫描App

移位寄存器型计数器

自启动

1. 环形计数器

2. 扭环形计数器 反馈 $D_0 = F(Q_0, \dots, Q_{n-1})$ 

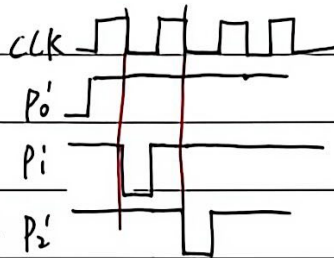
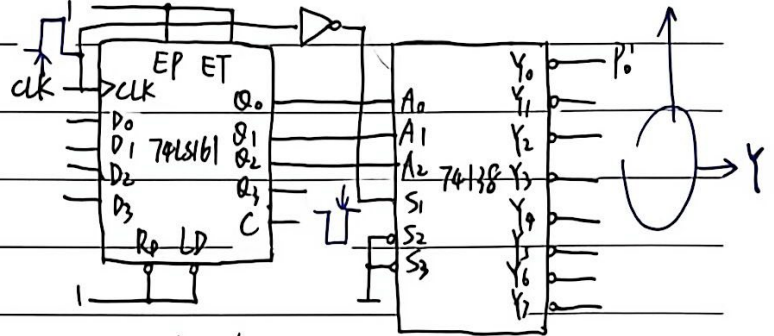
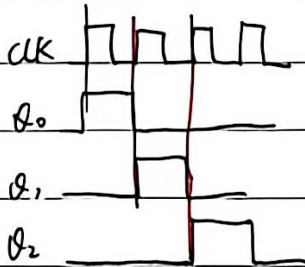
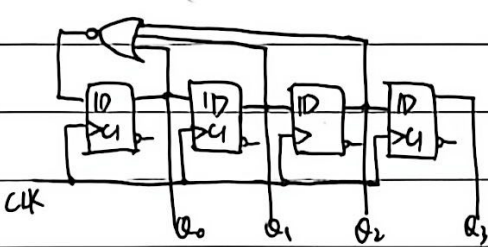
顺序脉冲发生器 空间多通道分发

序列信号发生器

① 用移位寄存器

② 用计数器 + 译码器

加组合电路变为



4位扭环形 + 有效循环: 消除竞争冒险



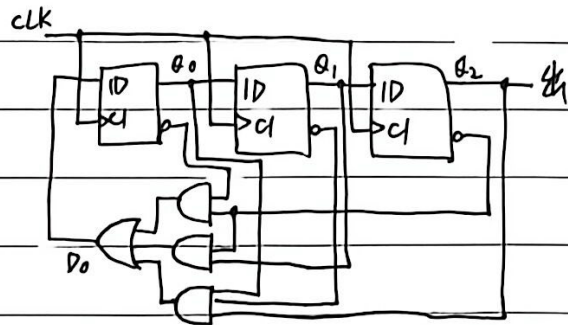
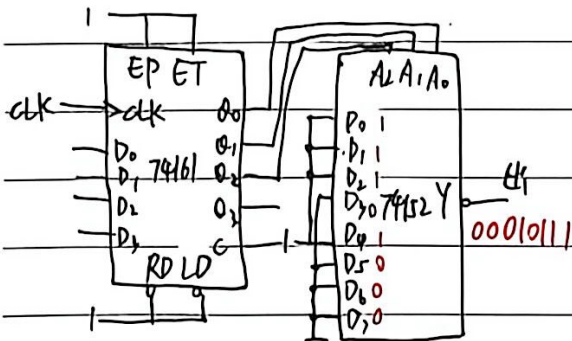
No.

Date. / /

序列信号发生器 单通道时间串行

① 计数器+数据选择器

② 带反馈逻辑的移位寄存器



CLK	Q ₂	Q ₁	Q ₀	Y'
0	0	0	0	0
1	0	0	1	0
2	0	1	0	0
3	0	1	1	1
4	1	0	0	0
5	1	0	1	1
6	1	1	0	1
7	1	1	1	1
8	0	0	0	0

CLK	Q ₂	Q ₁	Q ₀	D
0	0	0	0	0
1	0	0	1	0
2	0	1	0	0
3	1	0	1	1
4	0	1	1	0
5	1	1	1	1
6	1	1	0	1
7	1	0	0	1
8	0	0	0	0

$$D_0 = Q_2 Q_1' Q_0 + Q_2' Q_1 + Q_2' Q_0'$$



CS 扫描全能王

3亿人都在用的扫描App